

AI Astrobiology Life Detection & Biosignatures

NEW LITERATURE ADDED REGULARLY

2024:

Automated Discovery of Anomalous Features in Ultralarge Planetary Remote-Sensing Datasets Using Variational Autoencoders. 2024, Lesnikowski et al., *IEEE*

<https://ieeexplore.ieee.org/document/10444013>

Keywords: *Deep Learning, Generative Models, Technosignatures, Moon*

2023:

A robust, agnostic molecular biosignature based on machine learning. 2023, Cleaves H. J. et al., *PNAS*

<https://doi.org/10.1073/pnas.2307149120>

Keywords: *Random Forest, Biosignatures*

Signatures of Life Detected in Images of Rocks Using Neural Network Analysis Demonstrate New Potential for Searching for Biosignatures on the Surface of Mars. 2023, Corenblit et al., *Astrobiology Journal*.

<http://doi.org/10.1089/ast.2022.0034>

Keywords: *Biosignatures, Neural Net, CNN, Mars, Optimization*

Towards a Biosignatures Image Detection System for Planetary Exploration with UAVs. 2023, Galvez-Serna, J. et al. *IEEE Aerospace Conference*.

<https://ieeexplore.ieee.org/document/10115661>

Keywords: *Biosignatures, CNN, Imaging*

Using a neural network approach to accelerate disequilibrium chemistry calculations in exoplanet atmospheres. 2023, Hendrix, J. L. A. M et al. *Monthly Notices of the Royal Astronomical Society*.

<https://doi.org/10.1093/mnras/stad1763>

Keywords: *Exoplanets, Modeling, Neural Nets*

Orbit-to-ground framework to decode and predict biosignature patterns in terrestrial analogues. 2023, Warren-Rhodes, K. *Nat Astron.*

<https://doi.org/10.1038/s41550-022-01882-x>

Keywords: *Biosignatures, CNN, Deep Learning, Mapping*

2022:

Epistemology and anomaly detection in astrobiology. 2022, Kinney, D., Kempes, C. *Biol Philos.*

<https://doi.org/10.1007/s10539-022-09859-w>

Keywords: *Review, Fundamental Principles*

2006:

Acid Distribution in Meteorites: Diagenesis, Extraction Methods, and Standard Metrics in the Search for Extraterrestrial Biosignatures. 2006, McDonald G. D. and Storrie-Lombardi M. C., *Astrobiology Journal*.

<http://doi.org/10.1089/ast.2006.6.17>

Keywords: *Biosignatures, Cluster Analysis, Neural Nets*

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